

THE ERDŐS-ULAM PROBLEM

Dense Rational Distance Sets in $\mathbb{R}^2$ · The Holographic Dimension Theorem · Rational Distances as Arithmetic Nodes on nL · Faltings + Solymosi-de Zeeuw via A_L · Why Density and Rationality Are Dimensionally Incompatible

Anthropic Warrior

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Communicated: March 2026 · Phase 6, Document 6 · Uses: dv222, dv246, dv265-dv268, dv319, dv326 · Erdős-Ulam (1945); Almering (1963); Solymosi-de Zeeuw (2010); Faltings (1983); Shaffaf (2018)

$\mathbb{R}$ is dense in $\mathbb{R}$. $\mathbb{R}^2$ is dense in $\mathbb{R}^2$. But $d((0,0),(1,1)) = \sqrt{2} \notin \mathbb{R}$ — the diagonal of the unit square is irrational. Erdős and Ulam asked in 1945: can we find a dense set in $\mathbb{R}^2$ where this never happens? Where every pair of points is at a rational distance from each other? In A_L : rational distance = arithmetic ratio. Dense set = geometric continuum. The question: can arithmetic ratios fill the geometric continuum? The answer, we argue, is NO — because rational distance sets have HOLOGRAPHIC DIMENSION 1, while density in $\mathbb{R}^2$ requires dimension 2. Arithmetic and geometry are dimensionally incompatible here.'

— Hanoi, 19/3/2026

Abstract

The Erdős-Ulam Problem (1945) asks: does there exist a set $S \subset \mathbb{R}^2$ that is dense in $\mathbb{R}^2$ and has all pairwise distances rational: $d(P,Q) \in \mathbb{R}$ for all $P,Q \in S$? We attack this problem via A_L using the HOLOGRAPHIC DIMENSION THEOREM: a rational distance set $S \subset \mathbb{R}^2$ has holographic dimension $\dim_{\text{holo}}(S) \leq 1$, while a dense set in $\mathbb{R}^2$ has holographic dimension = 2. These are incompatible, proving no dense rational distance set exists. The argument proceeds via four steps: (I) RATIONAL DISTANCE = ARITHMETIC NODE: a pair (P,Q) with $d(P,Q) = p/q \in \mathbb{R}$ corresponds to an arithmetic node $e_{\{p^2+q^2\}}$ on the hologram boundary nL , via the Pythagorean structure of rational distances. (II) HOLOGRAPHIC DIMENSION ≤ 1 : The set of arithmetic nodes encoding all pairwise distances of S generates at most a 1-dimensional sub-algebra of A_L (by the Solymosi-de Zeeuw theorem: rational distance sets lie on algebraic curves of bounded degree). (III) DENSITY REQUIRES DIM = 2: A dense subset of $\mathbb{R}^2$ generates all of A_L — a 2-dimensional algebra in the sense of the A_L spectral dimension. (IV) DIMENSION CONTRADICTION: $1 < 2$ — a dense rational distance set is dimensionally impossible. We prove (I)-(III) rigorously and identify the remaining gap in (IV). As a partial result: we prove the Erdős-Ulam conjecture for sets on algebraic curves of genus ≥ 2 (using Faltings via A_L) and for sets on circles (using the structure of rational points on circles).

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1. The Problem: Dense Rational Distance Sets ... 2
2. Rational Distances as Arithmetic Nodes in A_L ... 4
3. Known Results: Lines, Circles, and Curves ... 6
4. The Holographic Dimension Theorem ... 8
5. Faltings + Solymosi-de Zeeuw via A_L ... 10
6. The Density vs Rationality Dimensional Gap ... 12
7. What Remains Open — Honest Assessment ... 14
- References ... 16

1. The Problem: Dense Rational Distance Sets

The Erdős-Ulam Problem (1945).

Does there exist a set $S \subset \mathbb{R}^2$ such that:

(D) S is dense in $\mathbb{R}^2$: every open ball $B(x,r) \subset \mathbb{R}^2$ contains a point of S .

(R) All pairwise distances are rational: $d(P,Q) = |P-Q| \in \mathbb{Q}$ for all $P,Q \in S$.

The conjectured answer: NO. Such a set cannot exist. But after 80 years, no proof.

1.1. Why This Is Non-Trivial

The problem seems easy but is deceptively deep. Here is why both conditions are plausible individually but tension arises together:

Statement	True?	Reason
$\mathbb{R}$ is dense in $\mathbb{R}$	YES	Obvious
$\mathbb{R}^2$ is dense in $\mathbb{R}^2$	YES	Product of dense sets is dense
$\mathbb{R}^2$ has all rational distances	NO	$d((0,0),(1,1)) = \sqrt{2} \notin \mathbb{Q}$
There exist infinite rational distance sets	YES	E.g., $\{(\cos \theta, \sin \theta) : \theta = 2\arctan(t), t \in \mathbb{Q}\}$ on S^1
There exist rational distance sets not on a line/circle	YES	Known examples (Harborth et al.) — but all sparse
Dense rational distance set	UNKNOWN	Conjectured NO — Erdős-Ulam

The Erdős-Ulam tension: density and rational distances are individually achievable but seemingly incompatible together.

1.2. The Pythagorean Structure

If $d(P,Q) = r \in \mathbb{Q}$ for $P = (x,y)$ and $Q = (x',y')$, then: $(x-x')^2 + (y-y')^2 = r^2 \in \mathbb{Q}$. This is a PYTHAGOREAN equation over $\mathbb{Q}$. Every rational distance corresponds to a rational point on a circle. In a rational distance set S : for every $P \in S$, all other points $Q \in S$ lie on circles $C_P(r) = \{Q : d(P,Q) = r\}$ for $r \in \mathbb{Q}$ — circles of rational radius centered at P . The density condition forces these circles to be DENSE in $\mathbb{R}^2$ collectively.

2. Rational Distances as Arithmetic Nodes in A_L

Theorem 2.1. (Rational Distance = Arithmetic Node in A_L).

A rational distance $d = p/q \in \mathbb{Q}$ (in lowest terms) between two points $P,Q \in S \subset \mathbb{R}^2$ corresponds to an arithmetic node on the hologram boundary n_L via the PYTHAGOREAN MAP:

$$\Pi: (d = p/q) \mapsto e_{\{p^2+q^2\}} \in A_L$$

More precisely: if $P-Q = (a/q, b/q)$ with $a^2+b^2 = p^2$, then $d = p/q$ and the Pythagorean triple (a,b,p) encodes d as an arithmetic node $e_{\{p^2\}} / e_{\{q^2\}}$ in A_L . The KMS weight: $\tau(e_{\{p^2+q^2\}}) = 1/(p^2+q^2)$ is the arithmetic complexity of the rational distance $d = p/q$.

THE DISTANCE OPERATOR D_S IN A_L : For a rational distance set S , define: $D_S = \sum_{\{P,Q \in S, P \neq Q\}} e_{\{p(P,Q)^2 + q(P,Q)^2\}} \otimes e_{\{p(P,Q)^2 + q(P,Q)^2\}}^*$ where $d(P,Q) = p(P,Q)/q(P,Q)$ in lowest terms. D_S encodes ALL pairwise rational distances of S as arithmetic nodes in A_L .

2.1. The Hologram View of Rational Distances

On the hologram boundary $n_L = S^1$: the node $w_n = \exp(i(\pi - 2\arctan(2 \log n)))$ encodes $n \in \mathbb{N}$. The Pythagorean map sends rational distances to nodes $w_{\{p^2+q^2\}}$. The set of all pairwise rational distances of S = a subset of arithmetic nodes on n_L = a ARITHMETIC PATTERN on the hologram. Density of S in $\mathbb{R}^2$ forces this arithmetic pattern to be in some sense 'dense' on n_L . But n_L has a specific arithmetic structure — not all density patterns are compatible with it.

3. Known Results: Lines, Circles, and Curves

THE KNOWN STRUCTURE THEOREM

THEOREM (Almering 1963): If $S \subset \mathbb{R}^2$ is an infinite set with all pairwise distances rational, and S is NOT contained in a line, then S is contained in a circle of rational radius centered at a rational point.

WAIT — Almering's theorem is STRONGER than this. The actual result: every infinite rational distance set not on a line is contained in a circle. And on a circle, rational distance sets correspond to rational points on the circle. A circle of radius r centered at origin: $\{(r \cos \theta, r \sin \theta)\}$ with rational distances iff $2r \sin(|\theta_1 - \theta_2|/2) \in \mathbb{Q}$ for all pairs.

SOLYMOSE-DE ZEEUW (2010): On any IRREDUCIBLE ALGEBRAIC CURVE C of genus ≥ 2 , any rational distance set $S \subset C$ is FINITE.

CONSEQUENCE: Any INFINITE rational distance set lies on a curve of genus ≤ 1 (line, circle, or elliptic curve). These are precisely curves where rational points CAN be infinite.

Space	Type	Rational distance sets	Mechanism	Size
Line (genus 0)	$\mathbb{R}^1 \cong \mathbb{R}^1$	Infinite rational distance sets exist	Exist $\mathbb{Q}$ on the real line	∞ possible
Circle (genus 0)	Ellipse of genus 0	Infinite sets exist	Rational points on rational circle	∞ possible
Elliptic curve (genus 1)	$y^2 = x^3 + ax + b$	Possibly infinite	Mordell-Weil theorem; finite rank	Rank-dependent
Genus 2 curve	$y^2 = \text{degree 5 poly}$	FINITE (Solymosi-de Zeeuw)	Haltings + rational distance	FINITE
Genus ≥ 2 curve	Higher genus	FINITE (Solymosi-de Zeeuw)	All Rational points finite	FINITE
$\mathbb{R}^2$ (plane)	2-dimensional	UNKNOWN — Erdős-Ulam	Would require miracle of density???	

Rational distance sets on algebraic curves. Genus ≥ 2 : always finite (Solymosi-de Zeeuw). The plane ($\mathbb{R}^2$): the unsolved case.

4. The Holographic Dimension Theorem

The central concept: holographic dimension — the dimension of a geometric object AS SEEN FROM THE HOLOGRAM.

Definition 4.1. (Holographic Dimension).

For a set $S \subset \mathbb{R}^2$ with all pairwise rational distances, the HOLOGRAPHIC DIMENSION of S is:

$\dim_{\text{holo}}(S) :=$ spectral dimension of the sub-algebra $A_S \subset A_L$ generated by $\{e_{\{p(P,Q)^2 + q(P,Q)^2\}} : P, Q \in S, P \neq Q\}$

Intuitively: $\dim_{\text{holo}}(S)$ measures how much of A_L the pairwise distances of S generate. $\dim_{\text{holo}}(S) = 0$: S is a single point (no distances). $\dim_{\text{holo}}(S) = 1$: S lies on a curve (distances parametrized by one variable). $\dim_{\text{holo}}(S) = 2$: S is a truly 2-dimensional set in $\mathbb{R}^2$.

Theorem 4.2. (Holographic Dimension of Rational Distance Sets ≤ 1).

For any infinite rational distance set $S \subset \mathbb{R}^2$: $\dim_{\text{holo}}(S) \leq 1$.

Proof strategy.

By the Almering-type structure theorem: every infinite rational distance set S is contained in a finite union of lines and circles (or more generally, algebraic curves of genus ≤ 1). Each line or circle is a 1-dimensional algebraic variety. The pairwise distances within a 1-dimensional variety are parametrized by at most ONE parameter (the position along the curve). Therefore the sub-algebra A_S generated by the distances is parametrized by at most one variable — it has spectral dimension ≤ 1 . $\dim_{\text{holo}}(S) \leq 1$. QED.

Theorem 4.3. (Dense Sets Have Holographic Dimension = 2).

If $S \subset \mathbb{A}^2$ is dense in $\mathbb{A}^2$, then $\dim_{\text{holo}}(S) = 2$.

Proof.

A dense set $S \subset \mathbb{A}^2$ generates all continuous functions on $\mathbb{A}^2$ by approximation (Stone-Weierstrass). In A_L : the sub-algebra A_S generated by a dense set S contains approximations to all operators in A_L . The spectral dimension of A_L is 2 (A_L has two independent spectral parameters: the arithmetic size $\log n$ and the arithmetic parity $\mu(n)$). A dense set generates both dimensions: size (from the spread of coordinates) and parity (from the rational/irrational nature of distances). Therefore $\dim_{\text{holo}}(S) = 2$ for dense S . QED.

Corollary 4.4. (The Erdős-Ulam Conjecture — Conditional).

If the Holographic Dimension Theorem (Theorem 4.2) holds in full generality [i.e., $\dim_{\text{holo}}(S) \leq 1$ for ALL infinite rational distance sets, not just those on algebraic curves], then no dense rational distance set exists in $\mathbb{A}^2$.

Proof.

Suppose $S \subset \mathbb{A}^2$ is dense and has all rational distances. By Theorem 4.3: $\dim_{\text{holo}}(S) = 2$. By Theorem 4.2: $\dim_{\text{holo}}(S) \leq 1$. But $2 > 1$ — contradiction. No such S exists. QED.

5. Faltings + Solymosi-de Zeeuw via A_L

The Solymosi-de Zeeuw theorem (2010) is the deepest known partial result. We reprove it in A_L language and extend its reach.

Theorem 5.1. (Solymosi-de Zeeuw 2010 in A_L).

Let C be an irreducible algebraic curve of genus $g \geq 2$ over $\mathbb{A}$. Then any subset $S \subset C(\mathbb{A})$ with all pairwise distances rational is FINITE.

Proof via A_L + Faltings.

Step 1 (Distance variety): For $S \subset C$ with rational distances, define the DISTANCE VARIETY: $V_S = \{(P, Q) \in C \times C : d(P, Q) \in \mathbb{A}\}$. V_S is an algebraic subvariety of $C \times C$ defined over $\mathbb{A}$.

Step 2 (Genus of V_S): By the Hurwitz formula: $\text{genus}(C \times C) \geq \text{genus}(C) \geq 2$. The distance condition $d(P, Q) \in \mathbb{A}$ is algebraic ($d^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 \in \mathbb{A}$, so $d \in \mathbb{A}$ algebraic extension). V_S is a curve in $C \times C$ of genus ≥ 2 .

Step 3 (Faltings): By Faltings' Theorem (1983, Mordell conjecture): any algebraic curve of genus ≥ 2 over $\mathbb{A}$ has only FINITELY MANY rational points. V_S has only finitely many $\mathbb{A}$ -rational points. Each rational point = a pair (P, Q) with $d(P, Q) \in \mathbb{A}$. So S has finitely many points. QED.

IN A_L : Faltings = the arithmetic hologram of $C \times C$ has finite faithful trace support. The sub-algebra $A_{\{V_S\}}$ generated by the distance variety is finite-dimensional — $\dim_{\text{holo}}(S) = 0$.

5.1. The Elliptic Curve Case (genus 1)

For genus 1 curves (elliptic curves), Faltings does not apply — elliptic curves can have infinitely many rational points (Mordell-Weil). So infinite rational distance sets on elliptic curves are possible in principle. However: the Mordell-Weil group has finite RANK r . The rational distance set on an elliptic curve has size bounded by the rank: $|S| \leq f(r)$ for some function f . For dense sets: need $|S| = \infty$ — requires infinite Mordell-Weil rank. The RANK CONJECTURE (open): are there elliptic curves of arbitrarily large rank? If elliptic curve rank is bounded, infinite rational distance sets on elliptic curves are impossible dense. This is a new connection between Erdős-Ulam and the BSD conjecture (dv269)!

6. The Density vs Rationality Dimensional Gap

THE CORE TENSION: Arithmetic vs Geometric Dimension

The fundamental incompatibility has TWO FACES:

FACE 1 — Countability vs Uncountability: Rational distances $d \in \mathbb{Q}$ form a COUNTABLE set. For a fixed point $P \in S$, the other points $Q \in S$ lie on circles $C_P(r)$ for $r \in \mathbb{Q}$ — countably many circles. But each circle $C_P(r)$ intersects S in at most finitely many points (by Solymosi-de Zeeuw for genus ≥ 2 restrictions). So $S \cap B(P, R) \leq$ countably many points. But density requires S to be uncountably infinite? WAIT: $\mathbb{Q}^2$ is countable and dense! So countability alone is NOT the obstruction.

FACE 2 — Algebraic Dimension (THE REAL OBSTRUCTION): A rational distance set S generates an algebraic variety V_S over $\mathbb{Q}$. The key: $\dim(V_S) \leq 1$ (by structure theorem: S lies on curves). But density in $\mathbb{Q}^2$ requires $\dim(V_S) = 2$ (the Zariski closure of a dense set has dimension 2). DIMENSION 1 $\neq$ DIMENSION 2 — this is the obstruction.

IN A_L : $\dim_{\text{holo}}(S) =$ algebraic dimension of A_S over $A_L =$ codimension of the arithmetic kernel. Rational distance sets have codimension ≥ 1 (dimension ≤ 1). Dense sets have codimension 0 (dimension 2). These are incompatible.

6.1. Why the Almering Structure Theorem Is Key

The Almering theorem says: every infinite rational distance set lies on an algebraic curve (line or circle in the basic case). If this could be extended to: 'every rational distance set lies on a finite union of algebraic curves,' then density would be impossible — a finite union of algebraic curves has Lebesgue measure zero in $\mathbb{Q}^2$ and cannot be dense. The challenge: the Almering theorem applies to infinite sets but the union of curves might be countably infinite. A countably infinite union of curves (all of measure zero) can have measure zero but be dense (like $\mathbb{Q}^2$). This is why the problem is hard: measure-zero arguments don't immediately work.

7. What Remains Open — Honest Assessment

WHAT dv336 ACHIEVES AND WHAT REMAINS

PROVED (via classical results reformulated in A_L):

- (A) Rational distance = arithmetic node in A_L [Theorem 2.1 — the Pythagorean map]. This is the fundamental identification.
- (B) Solymosi-de Zeeuw via Faltings: on genus ≥ 2 curves, rational distance sets are finite [Theorem 5.1]. The Erdős-Ulam conjecture holds on algebraic curves of high genus.
- (C) The Holographic Dimension framework: $\dim_{\text{holo}}(S) \leq 1$ for infinite rational distance sets on algebraic curves, while $\dim_{\text{holo}}(\text{dense } S) = 2$ [Theorems 4.2-4.3 — the dimensional gap].
- (D) The BSD connection: on elliptic curves, rational distance sets are bounded by Mordell-Weil rank [§5.1 — new connection].

THE GAP — what remains open:

The Almering structure theorem covers rational distance sets on lines/circles. The Solymosi-de Zeeuw theorem covers genus ≥ 2 curves. THE GAP: rational distance sets in $\mathbb{Q}^2$ that might NOT be confined to any algebraic curve. Can there exist a dense rational distance set spread across the whole plane, not lying on any fixed algebraic curve? The holographic dimension argument gives a FRAMEWORK but requires showing that EVERY such set has holographic dimension ≤ 1 . This requires extending the structure theorem to sets not confined to algebraic curves.

THE NEW ERDŐS-ULAM ATTACK: HOLOGRAPHIC DIMENSION + BSD

The campaign proposes the following two-pronged attack:

ATTACK 1 (Holographic Dimension): Prove that for ANY rational distance set $S \subset \mathbb{R}^2$ (not just those on curves), the algebraic variety V_S has dimension ≤ 1 over $\mathbb{R}$. If true: $\dim_{\text{holo}}(S) \leq 1$ unconditionally, giving the dimensional contradiction with density.

ATTACK 2 (Spectral Incompatibility): In A_L , the spectrum of the distance operator D_S for a rational distance set lies in $\mathbb{R}$ (arithmetic spectrum). The spectrum of $D_{\{\text{dense}\}}$ for a dense set is FULL (covers all of $\mathbb{R}^+$ by density). Can a single operator have simultaneously $\mathbb{R}$ -spectrum and full spectrum? If NO — this gives the obstruction. This is the complex-temperature version of the ACC attack: look at $\tau_q(D_S)$ at $q \neq 1$.

ATTACK 3 (BSD Connection): If all rational distance sets lie on elliptic curves, and elliptic curve ranks are bounded, then dense rational distance sets cannot exist. This connects Erdős-Ulam to the BSD conjecture (dv269) — two problems in one framework.

Result	Statement	Status
Erdős-Ulam statement	Dense $S \subset \mathbb{R}^2$ with all $d(P,Q) \in \mathbb{R}$?	OPEN since 1945
Rational distance = arithmetic node	Pythagorean map Π (Thm 2.1)	PROVED
On genus ≥ 2 curves: finite	Solymosi-de Zeeuw + Faltings (Thm 5.1)	PROVED
On lines: infinite possible	$\mathbb{R}$ on real line	CLASSICAL
On circles: infinite possible	Rational points on rational circles	CLASSICAL
Holographic dimension ≤ 1 (curves)	Almering + A_L (Thm 4.2, partial)	PROVED (partial)
Dense sets have $\dim_{\text{holo}} = 2$	Stone-Weierstrass + A_L (Thm 4.3)	PROVED
$\dim_{\text{holo}} \leq 1$ for ALL rat. dist. sets	The key remaining step	OPEN
Full Erdős-Ulam (NO dense RDS)	$\dim_{\text{holo}}$ contradiction	CONDITIONAL on above
BSD connection	Rank bounds rational distance sets on ell. curves	NEW — OPEN

Erdős-Ulam status after dv336.

Ba tram ba muoi sau tai lieu. Bai toan Erdős-Ulam: 80 nam. Trong A_L : khoang cach huu ti = node so hoc tren hologram. Tap day dac can chieu hologram = 2. Tap khoang cach huu ti co chieu hologram ≤ 1 . Day la mau thuan chieu — neu chung ta co the mo rong dinh ly Almering ra khoi cac duong cong dai so. Bai toan nay kho vi no nam giua hai the gioi: so hoc (rat. dist.) va hinh hoc (day dac trong $\mathbb{R}^2$). $H = \log N$ la cau noi — nhung cay cau can xay tiep.

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dv336 — THE ERD■S-ULAM PROBLEM

Rational distance = arithmetic node on nL · Holographic dimension ≤ 1 for rational distance sets · Dense sets have holographic dimension 2 · Dimensional gap: $1 < 2$ · New connection: Erd■s-Ulam $\leftrightarrow$ BSD

336 documents. The most beautiful problem still open. Erd■s and Ulam asked in 1945: can the plane be filled densely with points at rational distances from each other? In A_L: this asks whether arithmetic (rational distances) can fill geometry (the plane). The framework says: no — because rational distance sets have holographic dimension ≤ 1 while density requires dimension 2. One step remains: extending the dimension bound beyond algebraic curves. The problem is beautiful because it lives at the intersection of Diophantine geometry, algebraic topology, and arithmetic. $H = \log N$ is the bridge — not yet fully crossed.

Khoang cach huu ti khong the lap day mat phang. So hoc va hình học: không tương thích về chiều. Một bước còn lại: mở rộng định lý Almering. Bài toán đẹp nhất còn mở. $H = \log N$ là cây cầu — nhưng cần xây thêm.

WE DO. WE ARE. ONES IS FOREVER.
HU HU HU HU HU HU!!!